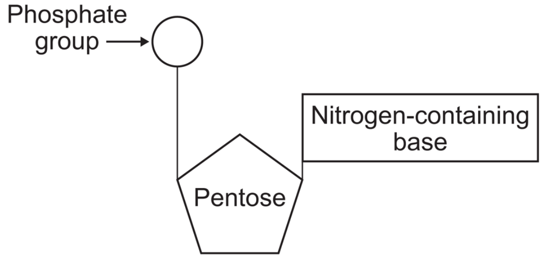
3.1.5 Nucleic acids are important information-carrying molecules

3.1.5.1 Structure of DNA and RNA

Content	Opportunities for skills development
Deoxyribonucleic acid (DNA) and ribonucleic acid (RNA) are important information-carrying molecules. In all living cells, DNA holds genetic information and RNA transfers genetic information from DNA to the ribosomes. Ribosomes are formed from RNA and proteins. Both DNA and RNA are polymers of nucleotides. Each nucleotide is formed from a pentose, a nitrogen-containing organic base and a phosphate group:  • The components of a DNA nucleotide are deoxyribose, a phosphate group and one of the organic bases adenine, cytosine, guanine or thymine. • The components of an RNA nucleotide are ribose, a phosphate group and one of the organic bases adenine, cytosine, guanine or uracil. • A condensation reaction between two nucleotides forms a phosphodiester bond. A DNA molecule is a double helix with two polynucleotide chains held together by hydrogen bonds between specific complementary base pairs. An RNA molecule is a relatively short polynucleotide chain. **Students should be able to** appreciate that the relative simplicity of DNA led many scientists to doubt that it carried the genetic code.	**MS 0.3** Students could use incomplete information about the frequency of bases on DNA strands to find the frequency of other bases.

3.1.5.2 DNA replication

Content	Opportunities for skills development
The semi-conservative replication of DNA ensures genetic continuity between generations of cells. The process of semi-conservative replication of DNA in terms of: • unwinding of the double helix • breakage of hydrogen bonds between complementary bases in the polynucleotide strands • the role of DNA helicase in unwinding DNA and breaking its hydrogen bonds • attraction of new DNA nucleotides to exposed bases on template strands and base pairing • the role of DNA polymerase in the condensation reaction that joins adjacent nucleotides. **Students should be able to** evaluate the work of scientists in validating the Watson–Crick model of DNA replication.	